

gretl working papers

Bayesian Averaging of Classical Estimates (BACE) for **gretl**

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Abstract

This paper presents a software package that implements Bayesian Averaging of Classical Estimates **BACE** ver. 1.3 for the **gretl** (the GNU regression, econometrics and time-series library).

Project's main page:

https://sourceforge.net/projects/bace-for-gretl

Warning

This PDF is outdated and needs to be updated.

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Changelog

1.3 Massive changes and improvements:

- (to doc) Correct model priors for truncated distributions.
- New MC3 strategy: taken directly from BMA package.
- New jointness measures.
- New `BACE()` bundle-based interface (default).
- Fix bug in `BACE_GUI()` in constant type selection.
- Fixed bug when „can be dropped" option is used.
- (to doc) New model types.
- (to doc) Updated docs.
- (to doc) Adding possibility for centering or standardizing variables.
- (do doc) New BIC versions.
- (do doc) Graphs added.
- (to doc) Overloading parameters in options bundle.
- Clean up the code.

1.2 Several changes:

- Some typos in output.
- Getting rid of calls to deprecated functions.
- New jointness measures.

1.1 Several changes:

- Documentation in PDF.
- Fix of several issues and bugs for model with only one explanatory variable and without constant.
- Change the option of the constant type from „random" to „can be dropped".

1.0 Initial version.

1 Introduction

In article Xavier Sala-i-Martin and Gernot Doppelhofer and Ronald I. Miller (2004) authors suggest a technique for averaging the parameter estimates – Bayesian Averaging of Classical Estimates (BACE), which enables to measure the importance of particular potential regressors. This approach is alternative to the familiar and earlier-applied Bayesian Model Averaging (BMA) technique, from which it differs mainly because of its less restrictive *prior* assumptions. The authors showed that

BACE weighting method may be understood as a limiting case of BMA approach when the *prior* information is dominated by the data. The averaging of the parameter estimates is performed across all possible combinations of linear regression models obtained by means of Ordinary Least Squares (OLS) with classical assumptions.

The BACE approach has several interesting features in relation to the techniques such as BMA or robustness-checking methods (reference). First of all, it is not necessary to make assumptions concerning *prior* distribution for parameters but only one simple assumption, on the expected model size.

An example of the application of the BACE technique for forecasting macroeconomic indicators can be found, among others, in the articles van Dijk (2004); Albis and Mapa (2014); Białowolski et al. (2014).

2 Bayesian Averaging of Classical Estimates

Consider simple regression model M_j:

$$y = X_j\beta_j + \epsilon \tag{1}$$

where y is $(Tx1)$ vector of observations, X_j is (Txk_j) matrix containing the set of independent variables and constant, β_j is a (k_jx1) vector of unknown parameters, ϵ is a $(Nx1)$ vector of errors which are assumed to be normally distributed i.e. $\epsilon \sim N(0_T, \sigma^2 I_T)$, $N(\mu, \Sigma)$ denotes a normal distribution with location μ and covariance Σ.

By applying Ordinary Least Squares (OLS) estimates (see Xavier Sala-i-Martin and Gernot Doppelhofer and Ronald I. Miller (2004)) we can calculate approximation of posterior probability of model M_j i.e. $\Pr(M_j|y)$ using the following formula: (add: Kass and Raftery (1995))

$$\Pr(M_j|y) \approx \frac{\Pr(M_j)T^{-k_j/2}SSE_j^{-T/2}}{\sum_{i=1}^{2^K}\Pr(M_i)T^{-k_i/2}SSE_i^{-T/2}}, \tag{2}$$

where SSE_j and SSE_i are the OLS sum of squared errors, 2^K denotes the total number of potential combinations of the K independent variables, k_j and k_i are the number of regression parameters β_j and β_i. Prior probabilities $\Pr(M_j)$ and $\Pr(M_i)$ of models M_j and M_i are binomially distributed, namely:

$$\Pr(M_j) = \theta^{k_i}(1-\theta)^{K-k_i}, \theta \in [0,1]. \tag{3}$$

Note that the Binomial distribution implies that we only need to specify a prior expected model size $\mathsf{E}(\Xi) = K\theta$ where $\mathsf{E}(\Xi) \in (0, K]$. If we define the value of $\mathsf{E}(\Xi)$ our BACE package will automatically produce a value of prior inclusion probability for all competitive models. In case of $\theta = 0.5$ prior expected model size is equal to the average of number of potential regressors, model prior distribution is uniform $\left(\Pr(M_i) = 2^{-K}\right)$ and reflects a lack of previous knowledge about models.

Add Beta-Binomial distribution (including truncated Atwood (1980); Tripathi et al. (1994))

Using BACE we can also easily evaluate the mean and variance of the posterior distribution of regression parameters β across the whole model space (see Leamer (1978)):

$$E(\beta|y) \approx \sum_{i=1}^{2^K} \Pr(M_i|y) E(\beta_i|y, M_i), \tag{4}$$

$$Var(\beta|y) \approx \sum_{i=1}^{2^K} \Pr(M_i|y) Var(\beta_i|y, M_i) + \sum_{i=1}^{2^K} \Pr(M_i|y) \left(E(\beta_i|y, M_i) - E(\beta|y)\right)^2, \tag{5}$$

where $E(\beta_i|y, M_i)$ and $Var(\beta_i|y, M_i)$ are the OLS estimates of β_i from model M_i.

Another useful characteristic popular in BACE approach is a Posterior Inclusion Probability (PIP). It is defined as the posterior probability that the variable x_i is relevant in explaining the dependent variable (see Leamer (1978); Mitchell and Beauchamp (1988)). In our case the PIP is calculated (using Monte Carlo integration) as the sum of the posterior model probabilities for all of the models including specific variable:

$$\Pr(\beta_i \neq 0|y) = \sum_{i=1}^{2^K} \Pr(M_i|y) \tag{6}$$

In many cases, despite of the interest of posterior characteristics it is valuable to predict future values of the dependent variable. In BACE approach we can carry out predictive inference across the whole model space. Given the matrix X_j^* containing explanatory variables for each of the out-of-sample data points we can predict inference on unobserved values of dependent variable as a weighted average of the individual model forecasts.

Doppelhofer and Weeks (2009)(DW2)

$$J_{DW2}(i,j) = \ln\left(\frac{p(i \cap j|y)p(\bar{i} \cap \bar{j}|y)}{p(i \cap \bar{j}|y)p(\bar{i} \cap j|y)}\right) \tag{7}$$

Ley and Steel (2007)(LS2)

$$J_{LS1}(i,j) = \ln\left(\frac{P(i \cap j|y)}{p(i|y) + p(j|y) - 2p(i \cap j|y)}\right) \tag{8}$$

Strachan Measure Strachan (2009)

$$J_{St}(i,j) = \; p(i|y)p(j|y) \ln\left(\frac{P(i \cap j|y)}{p(i \cap \bar{j}|y)p(\bar{i} \cap j|y)}\right) \tag{9}$$

Modified Yule's Q Measure (YQM) Hofmarcher et al. (2018)

$$J_{YQ}(i,j) = \ln\left(\frac{\left(p(i \cap j|y) + \frac{1}{2}\right)\left(p(\bar{i} \cap \bar{j}|y) + \frac{1}{2}\right) - \left(p(i \cap \bar{j}|y) + \frac{1}{2}\right)\left(p(\bar{i} \cap j|y) + \frac{1}{2}\right)}{\left(p(i \cap j|y) + \frac{1}{2}\right)\left(p(\bar{i} \cap \bar{j}|y) + \frac{1}{2}\right) + \left(p(i \cap \bar{j}|y) + \frac{1}{2}\right)\left(p(\bar{i} \cap j|y) + \frac{1}{2}\right) - \frac{1}{2}}\right) \tag{10}$$

A (B) indicates that event A (B) did not occur.

3 Usage of the BACE package

4 Implementation in gretl

In this section, we describe the code of the **BACE** package. First, we will characterize our code and the use of both: the command line and the GUI interfaces, then we will present some examples of using the **BACE**. Finally, we will show the outputs that are returned.

4.1 The core of the BACE code

The **BACE** package consists of 4 public and 14 private functions. The name of each function starts with the `BACE_` prefix. The public functions are:

1. `BACE()` see 4.1.1
2. `BACE_GUI()` see 4.1.2
3. `BACE_Print()` see **??**
4. `BACE_Plot()` see **??**

4.1.1 `BACE()`

This is a core package function which runs and controls the main loop. As a consequence, it is a prime interface to the **BACE** package. The `BACE()` function returns a bundle with all information needed to print and/or plot results anytime. One can print the results by calling `BACE_Print()` or plot graphs by calling `BACE_Plot()`. The signature of `BACE()` function is as follows:

```
function bundle BACE (series Y "Dependent variable",
 list X_list "List of indenpendent variables (X)",
 bundle Options[null])
```

The first two options are mandatory, while the last one is an optional collection of additional arguments. Those arguments are defined with the following default values:

```
scalar Options["Y_AR_p"] = 0
scalar Options["with_const"] = 1
scalar Options["model_prior"] = 2
scalar Options["avg_model_size"] = avg_model_size_tmp
scalar Options["alpha"] = 0.6
scalar Options["Nrank"] = xmin(2^(k+Y_AR_p_tmp-1), 40)
scalar Options["do_joint"] = 0
scalar Options["Npredict"] = 0
scalar Options["Nrep"] = Nrep
scalar Options["burn"] = 10
scalar Options["verbosity"] = 2
scalar Options["show_progress"] = do_show_progress
```

```
scalar Options["do_plot"] = do_plot_tmp
string Options["model_type"] = "linear"
scalar Options["data_trans"] = 0
scalar Options["bic_type"] = 1
scalar Options["hyperparameter"] = 0.5 # gamma for EBIC
scalar Options["binary_threshold"] = 0.5
scalar Options["_gui"] = 0
```

where k stands for the number of independent variables, `scalar avg_model_size` is calculated according to the following formula:

$$\texttt{avg_model_size} = \begin{cases} \texttt{(k + Y_AR_p_tmp)/2} & \text{if} \quad \texttt{with_const != 2}, \\ \texttt{(k + Y_AR_p_tmp + 1)/2} & \text{elsewhere}, \end{cases} \tag{11}$$

`scalar Nrep` takes the following values:

$$\texttt{Nrep} = \begin{cases} 10^3 & \text{if} \quad k \leq 10, \\ 10^4 & \text{if} \quad k \in [11; 15], \\ 10^5 & \text{if} \quad k \in [16; 19], \\ 10^6 & \text{if} \quad k \geq 20, \end{cases} \tag{12}$$

`scalar show_progress` takes the following values:

$$\texttt{show_progress} = \begin{cases} 1 & \text{if} \quad \texttt{\$sysinfo.gui_mode}, \\ 0 & \text{elsewhere}, \end{cases} \tag{13}$$

and `scalar do_plot` takes the following values:

$$\texttt{do_plot} = \begin{cases} 1 & \text{if} \quad \texttt{\$sysinfo.gui_mode}, \\ 0 & \text{elsewhere}. \end{cases} \tag{14}$$

Most of the optional parameters listed above are of numerical type and they may be gathered into following groups according to what kind of numbers they can take[1]:

1. Integers: `Y_AR_p`, `with_const`, `model_prior`, `Nrank`, `do_joint`, `Npredict`, `Nrep`, `burn`, `verbosity`, `data_trans`, `bic_type`.
2. Booleans: `show_progress`, `do_plot`.
3. Scalars: `avg_model_size`, `alpha`, `binary_threshold`.

Optional parameters control the following package options:

1. **Y_AR_p**: autoregression order of variable Y.
2. **with_const**: whether we include constant term in each estimated model?; possible values are: 0—without constant, 1—constant is always included, 2—constant may be dropped.

[1]Note, that the `scalar _gui` is for internal use only and must not be modified by user at any rate.

3. **model_prior**: model prior distribution to use; possible values are: 0—none, 1—Binomial, 2—Beta-Binomial.
4. **avg_model_size**: expected average model size for model prior.
5. **alpha**: significance level for the initial model.
6. **Nrank**: number of top ranked models.
7. **do_joint**: do we perform jointness analysis?; possible values are: 0—none, 1—Doppelhofer-Weeks Measure (DW1), 2—Doppelhofer-Weeks Measure (DW2), 3—Ley-Steel Measure (LS1), 4—Ley-Steel Measure (LS2), 5—Strachan Measure, 6—Yule's Q Measure (YQ), 7—Modified Yule's Q Measure (YQM).

4.1.2 BACE_GUI()

The `BACE_GUI()` is a wrapper to `BACE()` interface and takes the following parameters:

- **Dependent variable** as series [required].
- **Y lags** as integer with zero (only for time-series).
- **List of independent variables (X)** as list [may be null].
- **Constant in model** as integer [required]:
  1. Never – all models without constant,
  2. Always – all models with constant,
  3. Can be dropped – constant may be removed from or added to any model.
- **Model prior** as integer [required]:
  1. Binomial – binomial distribution,
  2. Beta-Binomial – beta-binomial distribution.
- **Prior average model size** as scalar with zero [required].
- **Significance level for the initial model** as scalar from $(0, 1)$ [required].
- **Number of the top ranked models** as integer [required].
- **Jointness analysis** as integer with zero [required][2]:
  1. None – we do not perform any jointness analysis,
  2. Ley-Steel Measure – we perform jointness analysis with Ley-Steel measure,

[2] Values ± 999.000000 should be interpreted as $\pm\infty$.

3. Doppelhofer-Weeks Measure – we perform jointness analysis with Doppelhofer-Weeks measure.

- **Number of out-of-sample forecasts** as integer with zero [required].
- **Total number of replications** as integer [required].
- **Percentage of burn-in draws** as integer from (0, 99) [required].
- **Verbosity** as integer [required].
- **Show overall progress** as Boolean.

4.2 The GUI way

Once you start `gretl`, you must open a data file and then you can load the relevant `BACE` package from the `gretl` server. In the main window, go to File → Function packages → On server heading. By selecting `BACE`, you will download the package on your local machine. Next, go to File → Function packages → On local machine and by selecting BACE from the list open a window similar to the one shown in Figure 1:

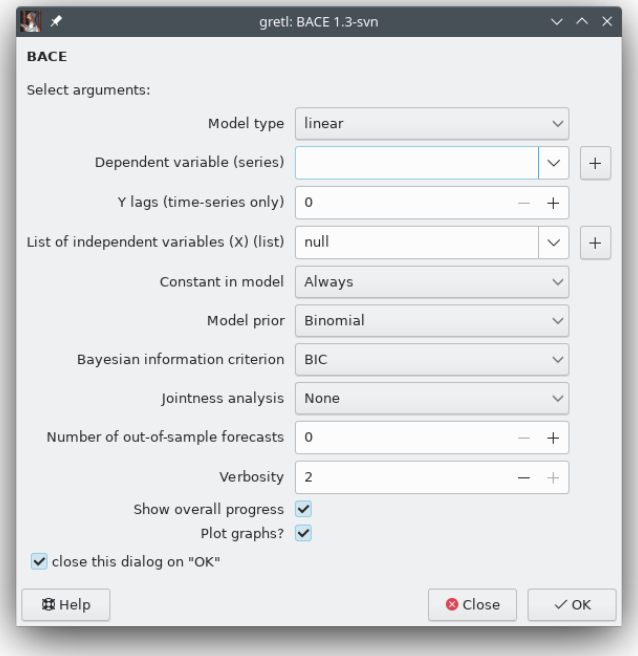


Figure 1: Main window for `BACE`.

According to Figure 1, we can specify the following entries in the GUI `BACE` window

- **Model type** – What kind of modeling should be should be performed?
- **Dependent variable** – The dependent variable.
- **Y lags** – number of lags for dependent variable (for time-series only).
- **List of independent variables** – List of all independent variables (including lagged series).
- **Constant in model** – Do we include intercept? It can be: always, never or can be dropped.
- **Bayesian information criterion** – Version of BIC to be used.
- **Jointness analysis** – If `None` (the default), the jointness analysis is omitted. Alternatively, one can choose one of the 7 available jointness measures.
- **Number of out-of-sample forecasts** – Defines the total number of out-of-sample forecasts of the dependent variable.
- **Verbosity** – An integer value of 1 or 2; the default is 2, so a more detailed description of the analysis is provided (initial model, speed of convergence, estimation results for top ranked models).
- **Show overall progress** – The progress bar is printed.
- **Plot graphs?** – Shall we produce graphs?

4.3 The script way

4.3.1 Script

The `BACE` package can also be used inside `Hansl` scripts. A very minimal code could be as follows:

```
open greene9_1.gdt
include BACE.gfn

list greene = capital labor nfirm

BACE(valadd, greene)
```

The above code triggers BACE analysis with default option described on page **??**. On contrary, the corresponding "big call", i.e. with setting all possible parameters, would be as follows:

```
bundle opts = defbundle("model_prior", 2, "avg_model_size", 1.5, \
  "alpha", 0.6, "Nrank", 4, "do_joint", 0, "Npredict", 0, \
  "Nrep", 1000, "burn", 10, "verbosity", 2, "show_progress", 1, \
  "do_plot", 1, "model_type", "linear", "const_type", 1, \
  "bic_type", 1, "data_trans", 0)

BACE(valadd, greene, opts)
```

or alternatively with the new syntax for creating a bundle on-the-fly:

```
BACE(valadd, greene, _(model_prior=2, avg_model_size=1.5, \
alpha=0.6, Nrank=4, do_joint=0, Npredict=0, Nrep=1000, \
burn=10, verbosity=2, show_progress=1, do_plot=1, \
model_type="linear", const_type=1, bic_type=1, data_trans
    =0))
```

Since some parameters in option bundle are over-loadable, the equivalent call looks like this:

```
BACE(valadd, greene, _(model_prior="beta-binomial", \
avg_model_size=1.5, alpha=0.6, Nrank=4, do_joint="none", \
Npredict=0, Nrep=1000, burn=10, verbosity=2, show_progress
    =1, \
do_plot=1, model_type="linear", const_type="always", \
bic_type="bic", data_trans="none"))
```

The above example consists of three blocks.

```
open greene9_1.gdt
include BACE.gfn
set seed 1000000

list greene = capital labor nfirm

scalar model_type = 1
scalar y_lags = 0
scalar with_const = 1
scalar data_trans = 0
scalar model_prior = 1
scalar bic_type = 1
scalar hyperparameter = 0
scalar prior_avg_model_size = 1.5
scalar alpha = 0.6
scalar top_ranked_model = 4
scalar joint = 1
scalar forecasts = 0
scalar Nrep = 2000
scalar Nburn = 10
scalar verbosity = 1
scalar progress = 1
scalar plots = 1

BACE_GUI(model_type, valadd, y_lags, greene, with_const,\
data_trans, model_prior, bic_type, hyperparameter,\
prior_avg_model_size, alpha, top_ranked_model, joint,\
forecasts, Nrep, Nburn, verbosity, progress, plots)
```

The above example consists of three blocks. The first block is just opening of the so called `greene9_1` dataset, which is bundled in every standard gretl installation. This dataset contains cross-sectional data on manufacturing of transportation equipment presented as Table 9.1 in Greene (1999)[3]. We also set the seed in order

[3]This dataset is also available at link http://people.stern.nyu.edu/wgreene/Text/Edition7/

to exact reproduction of the results.

The second block is the definition of the greene list which contains three independent variables available in the greene9_1 dataset.

The third block contains the definition for Bayesian Averaging of Classical Estimates analysis: no lags for Y variable (not applicable since we have cross-sectional data), constant always included, binomial model prior, prior average model size set to 1.5 (which means that we have uniform model prior), significance level for the initial model set to 0.6, 4 top ranked models, jointness analysis with Ley-Steel Measure, without out-of-sample predictions, 10000 replications with 10% burn-in draws, basic output (verbosity set to 1) and progress overall progress is printed.

Suppose we want to run the same analysis but following Occam's razor principle. The simplest way is setting the prior average model size to 1. We also want to have "random" constant, i.e. intercept can be dropped in any model. We also want to save results of the BACE analysis in the bundle results_b, the code should be as follows:

```
prior_avg_model_size = 1
with_const = 2

results_b = BACE_GUI(valadd, y_lags, greene, with_const, model_prior,
prior_avg_model_size, alpha, top_ranked_model, joint, forecasts,
Nrep, Nburn, verbosity, progress)
```

Suppose we want to calculate Doppelhofer-Weeks Measure instead of already computed Ley-Steel Measure. Since the type of jointness measure is controlled by do_join field in the result bundle, the only thing we have to do is to set the value of do_join to 2 and print BACE results again. The code should be as follows:

```
results_b.do_joint = 2
BACE_Print(&results_b)
```

4.4 Output

Outdated!!!

If you select the appropriate entries in the GUI BACE window, our package returns: Posterior Inclusion Probabilities (PIP), the averages of the posterior means and standard deviations of parameters ("Mean" and "Std.Dev.", respectively) and the posterior mean and standard deviation of each coefficient conditional on the variable being included in the model ("Cond.Mean" and "Cond.Std.Dev."). Furthermore, the package also returns the predictive results for the dependent variable ("Mean" and "Std.Dev.").

Let us consider the data used in previous example (australia.gdt). This dataset contains the Australian macro data used in Johansen (1995) and originally supplied by Tony Hall. The variables used in the text are:

TableF7-2.txt.

- lpau – log of Australian CPI (dependent variable),
- lpus – log of US CPI,
- le – log of exchange rate (price of US dollars in Aus. dollars),
- iau2 – 5-year Treasury bond rate, Australia,
- ius2 – 5-year Treasury bond rate, USA.

Suppose we want to perform BACE analysis up to $ADL(2,2)$ specification, i.e. all possible combinations from $ADL(0,0)$ to $ADL(2,2)$. The example script would be as follows:

```
set verbose off
include BACE.gfn
open australia.gdt
genr time

# We create lagged values
adl_lags = 2
lags(adl_lags, ius2)
lags(adl_lags, iau2)
lags(adl_lags, e2)
lags(adl_lags, lpus)
lags(adl_lags, le)

# We define list with all X-es (including lags)
list X_list = time ius2 ius2_* iau2 iau2_* e2 e2_* lpus lpus_* le le_*

set seed 1000000
scalar y_lags = 2
scalar with_const = 2
scalar model_prior = 1
scalar prior_avg_model_size = (nelem(X_list) + y_lags + (with_const > 0)) / 2
scalar alpha = 0.6
scalar top_ranked_model = 4
scalar joint = 0
scalar forecasts = 4
scalar Nrep = 20000
scalar Nburn = 10
scalar verbosity = 2
scalar progress = 0

RES = BACE_GUI(lpau, y_lags, X_list, with_const, model_prior, prior_avg_model_
alpha, top_ranked_model, joint, forecasts, Nrep, Nburn, verbosity, progress)
```

In the above analysis all variables (including dependent variable $lpau_t$) are lagged by 2 periods. Additionally we compute 4 forecasts:

```
Posterior moments (unconditional and conditional on inclusion):
PIP         Mean     Std.Dev.    Cond.Mean Cond.Std.Dev
lpau_1      1.000000     0.894595     0.091656     0.894595     0.09165
ius2      0.996500    -0.418560     0.102196    -0.420030     0.099314
lpus_2      0.790611    -0.303217     0.212152    -0.383523     0.16164
lpus_1      0.780500     0.356760     0.260342     0.457092     0.20243
iau2      0.752222     0.224487     0.161678     0.298432     0.112617
le_2      0.506944    -0.027670     0.046149    -0.054582     0.052271
const      0.392778    -0.074241     0.125663    -0.189015     0.136050
e2_2      0.335667     0.002317     0.033496     0.006903     0.057540
lpus      0.296222     0.037894     0.107134     0.127924     0.165014
le_1      0.282500    -0.023085     0.062683    -0.081716     0.095485
lpau_2      0.253056     0.037914     0.087083     0.149825     0.11489
e2_1      0.245222     0.015275     0.052532     0.062292     0.091240
time      0.233778     0.000053     0.000269     0.000226     0.000520
iau2_1      0.231389     0.035210     0.090726     0.152166     0.13332
le      0.211667     0.010063     0.042201     0.047542     0.081437
e2      0.176222    -0.005057     0.034623    -0.028698     0.078258
ius2_1      0.175222     0.021296     0.072659     0.121539     0.13396
ius2_2      0.140111    -0.009023     0.048048    -0.064400     0.11362
iau2_2      0.132000     0.005342     0.042150     0.040469     0.10971

---------------------------------

Posterior probability of models:
Model 1:        0.024278
Model 2:        0.020556
Model 3:        0.012889
Model 4:        0.009278
Total probability of the models in ranking (numerical): 0.067000

Correlation coefficient between the analytical
and numerical probabilities of the above models: 0.932557

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Predictive results:
lpau           Mean       Std.Dev.
1990:2        5.334649       5.340552       0.007950
1990:3        5.341856       5.356902       0.008400
1990:4        5.368310       5.378660       0.010094
1991:1        5.366443       5.395817       0.013705
```

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